

Working Principle & System Flow

How the Scuba Diver Health Monitoring System operates step by step

System Overview

The system continuously monitors the scuba diver's vital signs — heart rate, blood oxygen saturation (SpO2), and body temperature — using wearable sensors. Data is processed by the ESP32 transmitter and sent wirelessly through Li-Fi (light-based communication) to the surface team. If any parameter exceeds safe limits, the system automatically activates an emergency response and alerts the surface team in real time.

Step-by-Step Working

Step 1 — Power ON & Initialize	Both ESP32 units (Transmitter + Receiver) power up. Sensors initialize. MAX30102 and temperature sensors start reading data.
Step 2 — Continuous Sensor Reading	The ESP32 reads: Heart rate (BPM) from MAX30102, SpO2 (%) from pulse oximeter, Body temperature from DS18B20.
Step 3 — Data Processing & Emergency Check	ESP32 processes sensor values, checks against safety thresholds. Data is encoded into a digital signal for Li-Fi transmission.
Step 4 — Li-Fi Optical Transmission	The LED flashes data as light signals underwater. LED ON = binary 1 LED OFF = binary 0. Transmitted signal is captured by the surface team.
Step 5 — Li-Fi Reception & Decoding	Photodiode/LDR on surface captures light pulses. LM358 amplifier boosts the signal. ESP32 Receiver decodes the data.
Step 6a — Normal Condition	Heart rate: 60–100 BPM SpO2: >95% Temp: 36–37.5°C. LCD displays all values. No buzzer. Surface team receives normal status.
Step 6b — Abnormal / Emergency Condition	If heart rate < 50 or > 120 BPM OR SpO2 < 90% OR temp < 35°C or > 39°C: Surface buzzer activates. Red light flashes on the diver's wristband.
Step 7 — Panic Switch (Manual Alert)	If diver presses the panic switch: Immediate HIGH-PRIORITY alert sent via Li-Fi. Surface buzzer sounds continuous alarm.
Step 8 — Cloud Dashboard	All sensor data simultaneously uploaded to ThingSpeak cloud via ESP32 Wi-Fi. Real-time graphs and alerts available on the surface team's dashboard.

Health Parameter Safety Thresholds

Parameter	Sensor	Normal Range	Alert Condition	System Response
Heart Rate	MAX30102	60–100 BPM	<50 or >120 BPM	Buzzer + Relay + Balloon
SpO2	MAX30102	>95%	<90%	Buzzer + Relay + Balloon
Body Temperature	DS18B20	36–37.5°C	<35°C or >39°C	Buzzer + Alert
Panic Switch	Push Button	Not pressed	Pressed by diver	Immediate SOS alert